

Agentic Data Pipeline Orchestration for NASA Reanalyses — Colorado Snowpack

MERRA-2 and ERA5, Water Years 1981–2026

Built with coding agents and agent skills

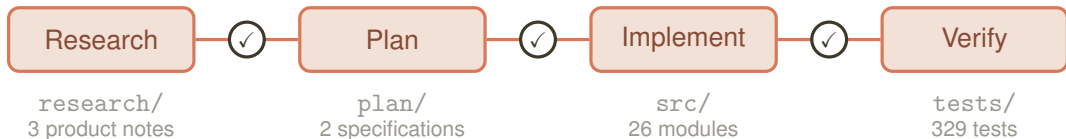
Responsible Gen-AI for NASA Earthdata Hackweek

responsible-genai.hackweek.io

August 28, 2026



Project Planning



Fixed Before Any Code

- the question
- domain and period
- variables, units, conversions
- the weighting rule
- what is out of scope

Fixed Before Any Number

- which years were extreme, not how much snow
- which model each number came from
- daily data for peak snowpack, never weekly
- not MERRA-2 in April, it has melted out
- where WY2026 rests on preliminary ERA5T

Agent Skills

SKILL.md

name:

description: *when to use it*

the traps

the conventions

what may be claimed

the agent matches on the description

... so it loads itself,
before it can get it wrong



Science

- reanalysis-snowpack-comparison
- snow-hydrology-fsca-evaluation
- modis-merra-regridding
- snow-bias-statistics-and-figures

Run Integrity

- earthdata-streaming-checkpoints
- resumable-fetch-run-integrity
- derived-quantity-integrity

Communication

- designing-thematic-figures

Eight skills, written during the work, from what went wrong in it.

The Data Pipeline

MERRA-2 Cloud OPeNDAP • earthaccess

ERA5 ARCO-ERA5 Zarr • Google Cloud

MODSCAG NSIDC DAAC • Earthdata Login

SNOTEL NRCS AWDB REST • WTEQ

3DEP USGS elevation • staged once



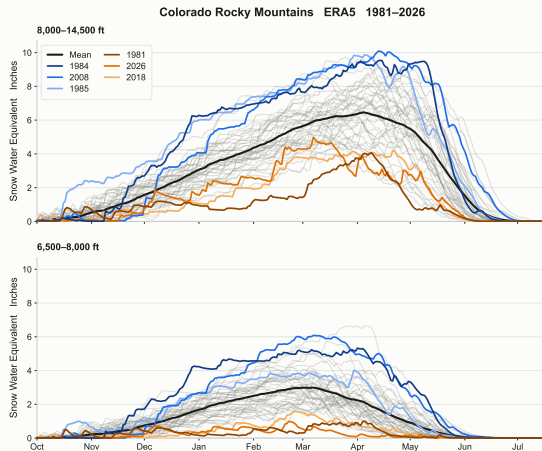
*resumable, and the whole report rebuilds
from the checkpoints with no network*

One Silent Trap per Archive

- **ERA5** *sd* is snow *water equivalent*, not depth. Geometric depth is sd / rsn
- **MERRA-2** *SNODP* is depth *within* the snow-covered fraction; the grid mean is $SNODP \times FRSNO$
- **ARCO-ERA5** the time axis runs to 2050. Absent chunks read back as NaN, with no error
- **SNOTEL** every station sits above the domain's median cell. Timing only, never magnitude

Each returns a plausible wrong number rather than an error. Each is now written down in a skill.

ERA5 by Elevation Band



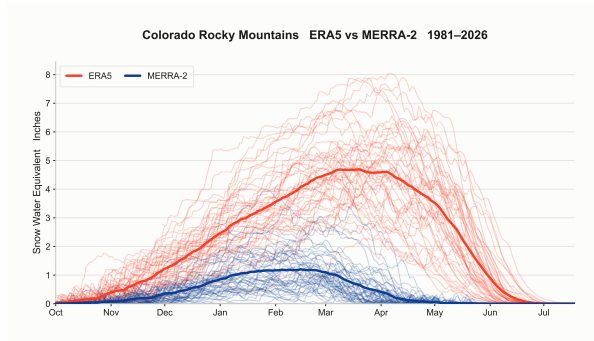
1 April SWE, per cent of band mean

	2023	2026
8,000–14,500 ft	140%	51%
6,500–8,000 ft	297%	5%

WY2026 is the lowest 1 April SWE of the 46 years, at 41% of the record mean. The deficit is not uniform: the lower band all but vanished.

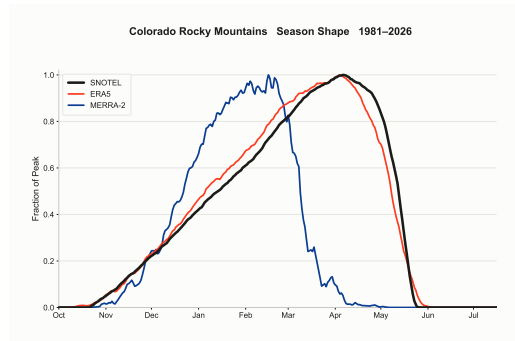
ERA5. The two models disagree in magnitude, so every number names its product.

Two Reanalyses, One Observational Check



A factor of 3.3 apart in mean peak SWE.
MERRA-2 exceeds ERA5's mean in 0 of 46 years.

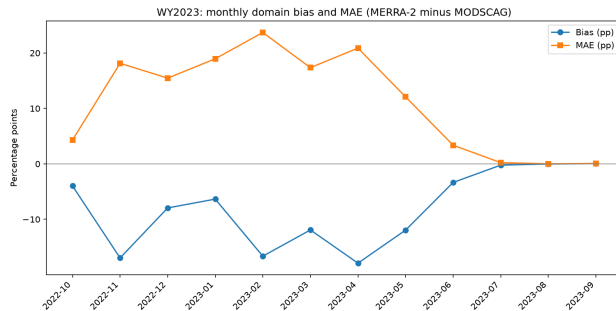
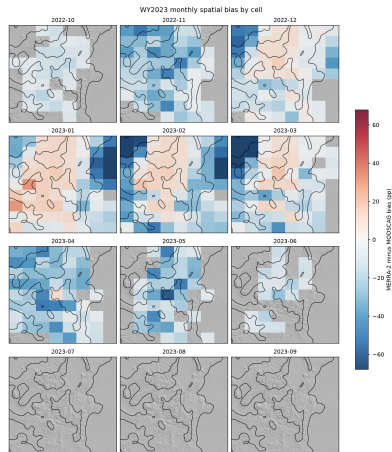
Yet they rank the years alike: $\rho = 0.73$



Each series scaled by its own peak.
SNOTEL arbitrates timing only.

MERRA-2 peaks two months early

fSCA Bias Month by Month, Water Year 2023

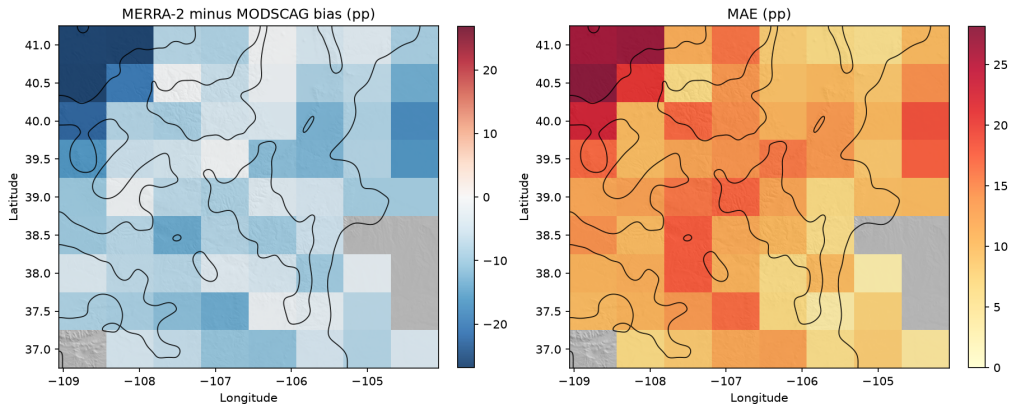


MERRA-2 minus MODSCAG

paired days 347 of 365
mean bias -4.3 pp
MAE 5.7 pp

The model runs low against the satellite through accumulation and converges at melt-out.

The Water Year as a Whole, Water Year 2023



Mean bias and MAE by cell, 347 paired days. Both largest in the northwest.

Melt-out, the last day cover stays above 0.10 — MODSCAG 20 May, ERA5 8 June, MERRA-2 26 April